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Review

Electric arc explosions—A review

Vytenis Babrauskas[a,b,*]

[a] Fire Science and Technology Inc., San Diego, CA, United States
[b] Dept. of Mechanical and Aerospace Engineering, University of California San Diego, La Jolla, CA, United States

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ABSTRACT

When an electric arc is created, a pressure event occurs. There can be two aspects to this: the shock and sound waves propagated from the expanding arc channel, and the bulk pressurization of the enclosure, if arcing is taking place within a closed volume. The present paper is the first systematic review of the research on both these pressure phenomena. Quantitative studies on electrical arc explosion pressures date back to the 1920s, although arc pressures generated by lightning, which is a type of electric arc discharge, have been studied since the 1700s, but understanding of the phenomena is still not complete or exhaustive. Experimental data are compared to theoretical predictions. It is shown that in an enclosed volume some extremely high pressures can be generated, if the arc current is sufficient. Such pressures can destroy buildings and mechanical equipment and cause injuries or death to nearby individuals. Even without enclosures, the shock waves produced from high energy arcs can cause injuries, although arc flash injury may be of greater concern. Injury potential generally requires that high currents be available, and serious damages or injuries are not associated with low-energy arcing occurrences.

1. Introduction

An electric arc is defined as a continuous, luminous discharge of electric current having a thermalized plasma and supported by thermionic emission from the cathode [1]. A plasma is a partially-ionized, electrically neutral gas, while 'thermalized' means that it is a hot plasma. Physically, an arc spans between two electrodes (metallic objects) maintained at a sufficient voltage difference. Geometrically, there is no standard shape for an arc. The shape of an arc is determined by the apparatus where it is created, and furthermore an arc typically changes its shape with time. The science of electric arcs is discussed in detail in numerous textbooks, e.g., Meek and Craggs [2].

Electric arcing in circuits with sizable maximum short-circuit current capacity can be a highly energetic effect. In fact, buildings have collapsed due to arc pressure, since in an enclosed space some surprisingly large pressures can be built up. 'Arc flash' is the thermal radiation component associated with energetic electric arcs, and it has received a great deal of study in recent years since thermal radiation has been a cause of serious burn injuries to electricians [3]. Consequently, computational methods and research have focused on the design of appropriate protective clothing. For arc pressures, however, no comparable activity has taken place. In fact, the literature is sparse and not systematic on this topic. It is the purpose of this paper to provide the first-ever review of electric arc explosions. The emphasis is placed on pressures developed and on the calculational methods available for these, along with experimental data that have been published. Some of the results are strikingly high. For instance, in one test explosion overpressures of 83 atm were obtained. The magnitude of this can best be appreciated by considering that a fuel-air deflagration will typically attain only around 7 – 8 atm, barring pressure-piling effects or other turbulence enhancements. During normal operation of a circuit breaker, arc pressures of roughly 3 atm magnitude can be expected [4], but these devices are designed to sustain the pressures generated by the normal arcing associated with circuit opening.

With regards to the energy supplying the arc, arcs can be of three types: (1) discharge of a fixed amount of stored energy (e.g., a capacitor; a current transformer); (2) DC power sources; or (3) AC power sources. Lightning strikes are the most important form of stored energy discharge, since capacitive discharge tends to be confined to specialized situations and is uncommon as a source of industrial accidents. Since most of the power transmission and distribution networks are AC, the bulk of the research available has focused on AC arcs. But heavy-power DC systems also exist and are important in certain industries (e.g., electric train propulsion). DC arc explosions are fundamentally different since there is no 'zero-crossing' in DC. In AC circuits, an arc will extinguish at 2× the power frequency (e.g., at 100 or 120 Hz), although it may reignite very shortly afterwards. In DC circuits, this extinguishing characteristic does not exist, and arcs will generally extinguish only due to external circuit interruption or due to

* Correspondence address: Fire Science and Technology Inc., San Diego, CA, United States.

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excessive electrode consumption. Most aspects of arc behavior only depend on arc current and arc power, and not on the type of power supply, but where the type of power supply does matter, this will be considered. The discharge of a fixed amount of energy is termed a *spark*, while a sustained discharge is an *arc*, but again this distinction will generally be seen not to affect the analysis of results.

Arc explosions are not rare in industry, and in other situations where 480 V, or higher, voltages are utilized, but published case histories are scarce. Neither of the two large electrical accident compilations [5,6] mentions the subject. Lee [7] published four brief case histories, Crawford et al. [8] documented seven case histories of arc explosions inside motor terminal boxes, include one fatality, while Heberlein et al. [9] described two non-fatal explosions inside motor control centers. The best-known incident was in an Atlanta high-rise building that took place on 30 June 1989. The fumbling of an electrician replacing a fuse caused a 480 VAC bus duct explosion [10] and the explosion and subsequent fire led to five fatalities.

Lightning strikes can lead to arc explosions in any type of premises. In 1773, Lind demonstrated that if a conductor from a lightning arrester is run down through a house, but with a small gap in this conductor, this can form a spark gap and a strike to the arrester can result in an arc explosion capable of destroying the house [11]. Individuals have been bodily knocked over when in proximity both to electrical fault arcs and lightning strikes, although interestingly often there have been negligible injuries to the individual knocked over [12]. But in cases where roofs collapse, the outcome may be traumatic if persons are present underneath.

Eardrum rupture can be expected at explosion overpressures of 19 kPa (10% probability) or 45 kPa (50% probability), while death due to lung damage is 120 kPa (10% probability) or 141 kPa (50% probability). The above values come from an extensive statistical study by Eisenberg et al. [13]; older data are somewhat different, but not greatly. In any case, they indicate that it does not take massive overpressures for injury or death to result from explosion pressures.

Only arc explosions in gases will be reviewed here, even though arc explosions in electrical insulating liquids can be of importance and arc explosions underwater are of some specialized interest. Apart from true arc explosions, conventional fuel-air explosions can also arise due to electrical causes. Perhaps the most common type of explosion associated with electricity is where an electric spark ignites a flammable gas mixture or dust cloud. These explosions can be severe and destructive, but are not covered here, because the electric power does not provide the energy for the explosion, instead, it is the chemical oxidation reaction which serves as the energy source. Also excluded are explosions in manholes, underground ducts, and similar installations which may entail both fuel-air explosion and arc explosion aspects. Transformer explosions [14] also involve some arc explosions aspects, but they are typically complicated and will likewise not be covered here.

An arc explosion arises due a very rapid heating of air or other medium. In the process, electrical energy is converted into other forms of energy: dissociation (breaking up of molecules, e.g., separating O_2 into 2O), ionization, and heating of the gas, including its compression; thermal radiation; and conduction losses into adjacent solids such as electrodes. In addition, some electrode metal is vaporized and this contributes to the total volume which is being explosively heated, yet, the role of chemical reactions has only recently been explored. Thunder is an acoustic manifestation of an electric arc explosion and has been studied for centuries (the currents involved in a lightning discharge are typically 20 – 200 kA, which is in the same range as for serious industrial electric arc accidents). Yet only in the last decade has there been some understanding achieved of the relative importance of the mechanisms involved, primarily due to Graneau and coworkers [15]. They demonstrated that thunder could not be adequately explained solely by thermal or electrodynamic effects, and this conclusion would pertain to arc explosions in general. In one experiment, they used an 8 μs duration capacitive discharge and showed that the energy measured from the arc was 124% of the energy stored in the capacitor. The 24% gain was not a measurement error, but represented chemical energy liberated by breaking chemical bonds. Specific pathways were found to include dissociating N_2 and O_2 molecules into atoms, then forming products such as O_3 and NO_X. Researchers have generally ignored the need to account for stored chemical energy, thus published heat balance calculations are likely to be subject to error, especially for short-duration arcs. Graneau et al. also noted that in the lightning community, high-current, short-duration discharges are termed 'cold lightning.' This is not hyperbole and they found that while discharges mechanically tore a piece of paper inserted into the path of the arc, they did not ignite the paper nor even char it. The paper could be ignited, however, if placed in contact with the electrodes. They also observed that lightning hitting water did not generate steam.

2. Shock waves generated from electric arcs

If in a gaseous medium there is an abrupt change in pressure, temperature, or volume created at some location, a wave will be generated which will propagate through the medium. The wave can be a sound wave, a shock wave, or both, depending on the characteristics of the source. In the case of an electric arc, while a shock wave will be generated and it is audibly perceived as an explosion (unless of very small scale), the shock wave does not constitute a detonation, which would require that the shock wave be supported by an exothermic reaction occurring behind the shock front.

For subsonic sound waves in air, the decay in pressure with distance from the source goes as $1/\sqrt{r}$ for the infinite-cylinder geometry and $1/r$ for the sphere. But for shock waves, these simple wave-equation relationships are not applicable. A reasonably short arc will be represented by a short cylinder, but this is not a geometry that lends itself to simple theoretical solutions. Baker presented calculated data on a point sources [16], along with experimental data on bursting explosions of short cylindrical vessels [17] and spherical vessels[17]. The curvature in these relationships (Fig. 1) show that, unless only examined over small intervals, the actual relationship is not of a power-law type. In this figure, the non-dimensional scaling variables used are $\overline{P} = \Delta p/p_o$ and $\overline{R} = r/\sqrt[3]{E/p_o}$, where Δp= pressure rise (Pa), p_o = ambient pressure (Pa), r = radius (m), and E = energy (J). But, in addition, it should be noted that the actual shape of the curve will depend on the rate at which the energy is delivered, and this can have a range of variations.

Turning now specifically to the electric arc, when breakdown is initiated, a narrow conducting filament first bridges the gap, and then it grows rapidly in diameter until it reaches an ultimate value and the 'arc

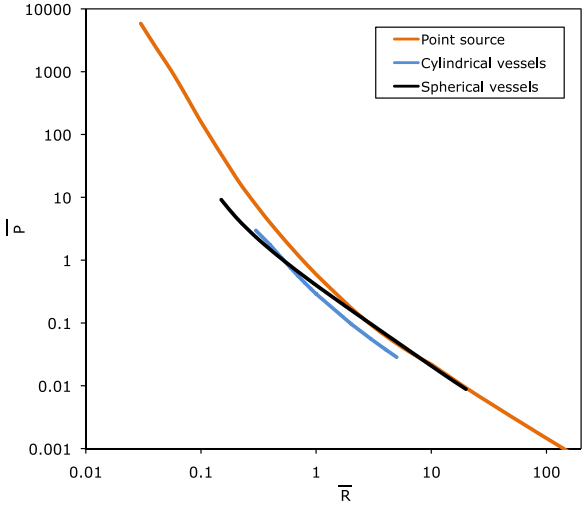


Fig. 1. Shock wave overpressure scaling for several geometries.

channel' is fully established. In 1936, Suits [18] measured pressures and sound waves emitted from the expanding arc channel and determined that there are two components: a sound wave propagating at the speed of sound and a shock wave propagating at 2 or 3 times that. He also documented the interesting phenomenon that a shock wave traveling through the air can often extinguish an ongoing arc, if it crosses its path. This work was extended by Flowers [19], who in 1943 made more detailed measurements and found velocities of 1000 – 2000 m s^{-1} for the radial expansion of the arc channel. His slowest measured rate of expansion was 460 m s^{-1}, which is still above the 340 m s^{-1} of sound speed, indicating that shock waves are being generated in all cases. The arc channel eventually reaches a steady-state diameter and no longer expands, and Flowers found that a time of 3 – 35 μs was required for the final diameter to be attained in his experiments, but the actual value is dependent on external circuit parameters which limit the current growth rate. When the final diameter is achieved, Flowers found that the arc channel cross-sectional area is linearly proportional to the current, with the proportionality being 11 A mm^{-2}.

Later Vanyukov et al. [20] showed that the expanding shock front and the channel are initially of the same diameter, but subsequently the channel typically approaches a maximum diameter, while the shock front continues expanding outwards (Fig. 2). Intermediate between these is the 'envelope,' which is the boundary between compressed gas (outside the boundary) and rarefied gas (inside). In some cases, however, arc channel diameter growth continues for a protracted period, especially at higher current values [21]. Shock wave modeling was not yet existent when Flowers did his study, and it was not until 1951 that Drabkina [22] presented results of shock wave modeling of an arc and showed that, for arcs supplied with a small current, the shock wave may only propagate for 25 mm, or so, before the shock magnitude becomes insignificant (Fig. 3). Eventually with sufficient distance, any shock wave dissipates into a sound wave. Drabkina's modeling results were found to be inconsistent with experiments of Koppitz [23], but then Hallgren and Sigmond [24] showed that this was due to an error in calculating two constants and, with that correction, her theory shows very good results with experiment. Her 1951 paper is remarkable since it comprised computational fluid dynamics (CFD) calculations before the availability of digital computers.

Later, Hill [25] modeled the shock wave development of a 30 kA arc with hydrodynamic calculations. Data were not available for validation of pressures, but Hill successfully validated the temperature profiles he computed against experimental data. Freeman and Craggs [26] further continued modeling the shock front development by considering various rates of energy delivery, as a function of time. Graneau [11] studied high-current arcs discharged across a small (3.2 mm) gap and

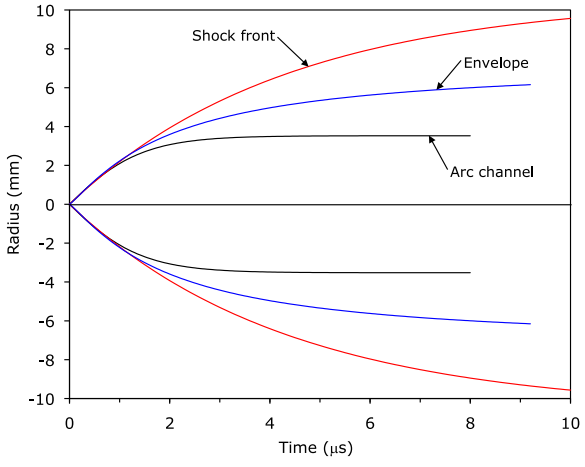


Fig. 2. Results of Vanyukov et al. [20] on the expansion of the arc channel and growth of the shock wave.

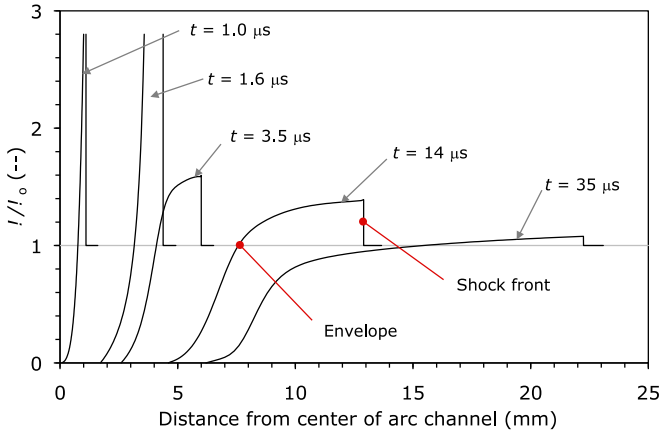


Fig. 3. Modeling result of Drabkina [22] on development of the arc channel.

found that the shock-front region was wedge-shaped and would not be well modeled as a spherical or a cylindrical source. Thus, a spherical representation cannot be assumed to be well suited under all conditions.

3. Pressure from arcs in the open

Two different conditions have to be considered with regards to pressures produced from arcs. If the arcing is in the open air, or in a large space (e.g., an open factory area), then there will not be a significant pressure rise in the whole volume, but there will be local pressure effects in the vicinity of the arc. Conversely, if arcing is an a switchgear cabinet, a small substation, or a similarly enclosed volume, then significant pressure rise can be experienced throughout this entire volume. The main research focus has been on closed or semi-closed volumes and information on arc pressures in the open is limited.

Baker [16] proposed that arc explosions in large enclosures or in the open can be treated using some results from acoustical theory. The pressure rise $\Delta p = (p_1 - p_o)$ is assumed to be due to the arc effectively generating a certain volume V of air at the ambient density ρ_o. Then from acoustical theory, the pressure rise at any particular distance r (m) from the arc is:

$$\Delta p = \frac{\rho_o \ddot{V}(t)}{4\pi r}$$

where $\ddot{V}(t)$= second derivative of the volume occupied by the arc (m^3 s^{-2}). If the pressure rise is small, then the Ideal Gas Law can be used in the form $p_o(V_1 - V_o) = nR(T_1 - T_o)$ where R = universal gas constant =8.314 J mol^{-1} K^{-1}. At constant pressure, the energy increase ΔE is given by $\Delta E = n\, c_p(T_1 - T_o)$, where n = number of moles, and c_p = constant-pressure heat capacity (J mol^{-1} K^{-1}). But $c_p = c_v + R$, where c_v = constant-volume heat capacity (J mol^{-1} K^{-1}). It becomes convenient to eliminate c_p and to use the adiabatic coefficient γ defined as:$\gamma = c_p/c_v$. Air can assumed to be an ideal diatomic gas, giving γ=7/5=1.4; while for sulfur hexafluoride, SF_6, which is often used in high-voltage switchgear applications, γ=1.122. Eliminating $(T_1 - T_o)$ from the above two expressions and solving for the volume change gives:$(V_1 - V_o) = \frac{\gamma - 1}{\gamma}\frac{\Delta E}{p_o}$

Taking the second derivative with respect to time gives:

$$\ddot{V}(t) = \frac{\gamma - 1}{\gamma}\frac{\Delta \ddot{E}}{p_o}$$

and the pressure rise is:

$$\Delta p = \frac{\gamma - 1}{\gamma}\frac{\rho_o \Delta \ddot{E}}{4\pi r p_o}$$

but, since $P = \dot{E}$, the expression can be given in terms of the rate of rise

of power (note that P here denotes power and not pressure)[1]:

$$\Delta p = \frac{\gamma - 1}{\gamma} \frac{\rho_o \dot{P}}{4\pi\, r p_o}$$

This equation predicts that the pressure rise will vary linearly with $\dot{P}$ and decrease with distance proportionally to $1/r$.

In 1969, Uman et al. [27] studied the shock waves produced from 3 MV impulse sparks discharging 20 kJ across a 4 m spark gap and showing a peak current of 10 kA. At the measurement station closest to the spark channel, 0.35 m, a maximum overpressure of 0.1 atm was measured, while at 4 m distance, the maximum was only 0.01 atm. An exceedingly long arc of this kind could be expected to be reasonably represented as an infinite cylinder. Plooster [28] modeled the infinitely-long cylindrical geometry and showed that, in the far field, pressure decays as $1/r^{3/4}$, but Uman's measurements did not bear this out. The basic problem is that long spark gaps tend to yield sparks which are sinuous or branched, so they are more appropriately treated as a series of shorter-segment sources, but applying theory to this type of geometry is difficult. In a branched spark, each segment should be treated as a separate source of a shock wave, and Uman found that the composite action of these multiple segments is much more closely approximated by Brode's theory of spherical explosions [29], with an effective sphere diameter of 0.5 m best representing their experimental data at all distances > ¼ of the actual 4 m gap length. Fig. 4 shows the experimental results along with predictions according to Brode's theory; the values reported are the overpressure (Δp) divided by the ambient pressure (p_o). Since in most cases of interest the pertinent arc gap distances will be «4 m, this suggests that practical cases should generally be treated as having a spherical source geometry. Uman also found that, to fit the theory, an effective energy of $^1/_6$ of the actual supplied energy needed to assumed, but did not endeavor to perform heat balance calculations to determine where the remaining $^5/_6$ was expended.

Drouet and Nadeau [30] measured the overpressures from 60 Hz AC arcs using horizontal arc gaps of 1.5 – 6 m, with voltages up to 18 kV and currents of 1 – 80 kA. The arc did not significantly change its shape during any single cycle of the waveform, but due to convective heating of the air, it progressively grew in length to 15 m after 0.5 s of arcing. They also conducted experiments with very short arcs, 8 mm long, using a 100 V power supply. In analyzing their data, the authors assumed that the pressure×distance product was invariant with distance, in other words, that the pressure shows a $1/r$ dependence. They also found that $p \times r$ was related to the time derivative of the arc power, $\frac{dP_a}{dt}$ or $\dot{P}_a$, where P_a denotes the electrical arc power. They then expressed this as: $\dot{P}_a = 2\pi f\, V_{rms} I_{rms}$, where f = frequency (Hz). For AC arcs, the effective frequency for power is twice that of the voltage or current waveform; thus, for a 60 Hz supply, f =120 Hz. This expression for $\dot{P}$ is mathematically correct only if the phase angle is 0, but is further misleading since it would imply that the pressure created by a DC arc is nil, whereas this is clearly untrue. Their experimental results showed two regimes. The high power arcs could be represented as:

$$\Delta p = \frac{1.0 \times 10^{-6}}{r} \dot{P}_a^{0.9}$$

where p= arc pressure (Pa), and r = radial distance from arc (m). For low power arcs, a relation that varies exactly linearly with $\dot{P}$ fits the data:

$$\Delta p = \frac{1.3 \times 10^{-6}}{r} \dot{P}_a$$

Note that, using γ=1.4 and p_o =101×10^3 Pa, Baker's theoretical expression would give

[1] Correcting for a typographical error in the original reference, where the density factor is missing.

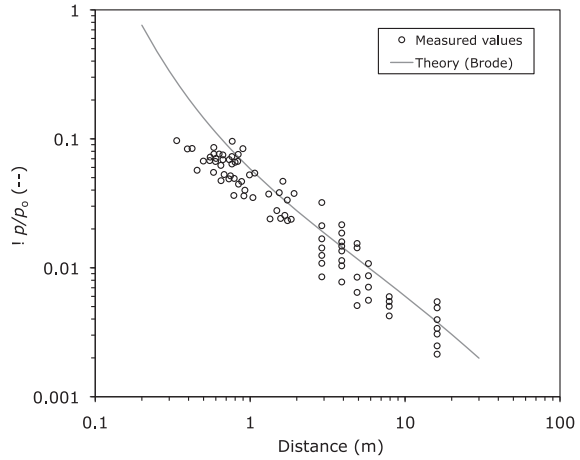


Fig. 4. Pressures measured by Uman [27] on a 4 m spark, along with Brode's theoretical model results[29].

$$\Delta p = \frac{0.270 \times 10^{-6}}{r} \dot{P}_a$$

Fig. 5 show a comparison of these relationships. Baker's theoretical solution is a reasonably good fit to the long arc results, except for the fact that some (expected) losses are seen and the power law is slightly below 1.0. The reason why the short arc results should be substantially above the theoretical prediction is not clear, but possibly this is due to measurement errors in the determination of $\dot{P}$.

Lee [7] referred to the work of Drouet and Nadeau and provided graphs which express the equation:

$$\Delta p = \frac{k\, I_{rms}}{r^{0.9}}$$

Lee's paper has been widely referenced, but the above relationship (Lee provided only the graphs and not an equation) is not a correct version for either regime in Drouet and Nadeau's data. Both voltage and the time derivative are missing from Lee's formulation, and the power-law for distance is also incorrect.

4. Pressure from arcs in an enclosure

If an arc discharge occurs within an enclosure which is of modest size, then the whole enclosure will get appreciably pressurized. In

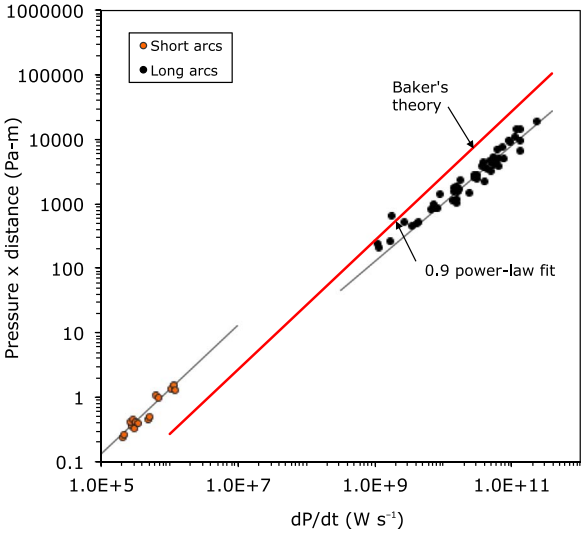


Fig. 5. The results of Drouet and Nadeau [30] for short (8 mm) and long (many meters) arcs.

addition, of course, there will be local shock wave propagation, and also reflections of shock waves from compartment walls. Modeling such details in the system would require numerical calculations. But, for many practical purposes, what is of most interest is the peak, quasi-steady overpressure that is achieved, and this can be approximated in a simple way. Neglecting all transient and hydrodynamic effects, the discharge of an arc in a single, closed compartment can be treated as an ideal gas within a isolated, isochoric (constant volume) system. If an amount of heat or energy ΔQ is injected into the volume, the change in pressure Δp is [29,31–33]:

$$\Delta p = \frac{R}{MC_v}\frac{\Delta Q}{V}$$

where M = molar mass, and the constant-volume heat capacity, C_v, here has the units J kg^{-1} K^{-1}. Using the relations between C_v, C_p, and γ, the relation is more usefully written as:

$$\Delta p = (\gamma - 1)\frac{\Delta Q}{V}$$

A number of investigators have conducted tests in enclosures of various sizes and reported their findings. Consistent with the theoretical prediction, when using very small enclosures and large arc currents, some exceedingly large pressures can be obtained. Graneau [11] conducted experiments in a tiny cubical cavity, 12.7 mm on a side (2×10^{-6} m^3 volume), with the electrode gap also being 12.7 mm. For a spark discharge of 40 kV and a peak current of 38 kA, he measured an average pressure of 409 atm in the cavity, with even higher pressures during the peak of the discharge.

Slightly larger enclosures were employed by Allsop and Wheeler [34], who conducted the first systematic study of arc pressures in 1927. With their original test enclosures (0.1×10^{-3} – 0.9×10^{-3} m^3), they found that as the size of the enclosure decreased, the pressure *decreased*. This obviously suggests that simple application of the Ideal Gas Law is not sufficient to describe the physics involved. For modest DC arcs (up to 3 kJ) in a small enclosure, the authors concluded that a major effect was heat loss due to condensing of vaporized copper on the enclosure walls—the smaller the enclosure, the higher the fraction of energy lost by this mechanism. In this test environment, they also did *not* get a linear arc energy vs. pressure relationship. Instead, for increasing arc energies, the pressure rose less than linearly. Again, this is suggestive of a dominant effect of enclosure walls, with progressively greater loss of heat at higher arc energies. For an enclosure of 0.9×10^{-3} m^3 and an arc energy of 1 kJ, the authors obtained an overpressure of 130 kPa, while for a 3 kJ arc, they obtained an overpressure of 200 kPa. To evaluate significantly lossy systems, in the above theoretical relationship it is appropriate to replace ΔQ with $k_p\Delta Q$, where k_p = fraction of energy going into raising the gas pressure. Then for the 1 kJ arc used, k_p =0.29, while for the 3 kJ arc k_p =0.15. This indeed indicates that small enclosures are highly lossy. The authors also endeavored to repeat their results with AC arcs, but found that, with comparable circuit parameters, the arc energies and pressures were so much lower than with DC that they could not get reliable data.

In their next paper, Allsop and Wheeler [35] reported on further experiments with DC arcs, giving arc current, voltage, and power data, but these would not have general applicability since, in these experiments, they used an arrangement where the electrodes were continuously moving apart. For a 125 kW arc with a 527 V power supply, they obtained a peak overpressure of 8.9 atm. In the third paper [36], Allsop tested copper fuse links in a 6.6×10^{-3} m^3 enclosure. At 8 kJ, a pressure of 170 kPa was registered, and for lower arc energies there appeared to be a linear relationship. But at higher arc energies, the pressure increase was proportionally less. At an energy of 14 kJ, an overpressure of 230 kPa was found, while at 120 kJ energy 680 kPa resulted. In terms of the k_p factor, this gives k_p =0.35, k_p =0.27, and k_p =0.08. Between 120 and 320 kJ arc energy, there was essentially no increase in arc pressure. These arcs were self-clearing, since the fuse link that melted created a gap longer than could sustain the arc. In the third study, medium-small enclosures (2.5×10^{-3} to 6.7×10^{-3} m^3) were used, and for these sizes the pressure did not show a dependence on the volume, suggesting that the Ideal Gas Law effect and the wall heat loss effects essentially canceled out. In the final study, Allsop [37] found out that in oxygen, lower arc pressures get developed than in air. This was determined to be a chemical effect, in that an oxygen atmosphere promotes reacting copper vapors to nonconductive copper oxide and helping to reduce potentially conductive copper paths being established on nearby surfaces. This effect would presumably be smaller for enclosures more sizable than the tiny ones (0.7×10^{-3} m^3) studied by Allsop.

Hennecke and Horn [38] conducted arc pressure tests in a 0.63 m^3 cylindrical steel vessel using a 400 VAC, 50 Hz power source. They found that the overpressure increased nearly linearly with duration of the arc up to ca. 150 ms. The pressures obtained were 10 – 20% higher when aluminum electrodes were used instead of copper. They performed a rough heat balance and concluded that for copper electrodes around 50% of the arc power went into heating and pressurizing the gas, while with aluminum electrodes this rose to 65%. About 10% represented losses to heating and vaporizing electrode material, while the remaining comprised radiation, conduction, and convection losses.

Baldrey and Hudson [33] conducted tests within a small pressure vessel using a 3800 V_{rms} power source, a 50 mm electrode spacing, and an available AC short-circuit current I_{sc} =8000 A. The pressure vessel was protected by using burst discs of various ratings, and it proved readily feasible to destroy discs rated at 7 atm. They then conducted large-scale tests where fault currents up to 45 kA were available. Much greater overpressures were recorded, being 83 atm in the worst case.

Heberlein et al. [9] conducted arcing tests inside 600 VAC motor control center boxes of unspecified volume and attempted to correlate the measured overpressures to the I^2t of the arc. In 9 of 10 tests the doors were not blown open, but no correlation emerged. The peak overpressure found was 0.75 atm, for an arc with 4.5×10^6 A^2-s, but an arc with an I^2t some 6.9 times greater showed an overpressure of only 0.31 atm.

The largest body of research on arc pressures in enclosures has been that of Pietsch and coworkers at the RWTH Aachen University. In one study [32], they used capacitive discharges between copper, aluminum, or steel electrodes in an 0.8 m^3 enclosure. These tests showed a linear relationship between arc energy and pressure rise in a closed compartment (Fig. 6). The measurements showed that 40% of the arc energy was recovered as a pressure rise within the gas when using aluminum or copper electrodes. Other tests, details of which were not published, indicated that when using steel electrodes only 25% of the arc energy was recovered as a pressure rise. The Aachen researchers analyzed most of their data using a steady-state heat balance [39]:

$$P_{EI} + P_{chem} = P_p + P_r + P_m$$

where P_{EI} = electric power of the arc, P_{chem} = power generated due to chemical or ionization reactions, P_p = power going into heating and raising the pressure of the gas volume, P_r = power radiated away, and P_m = power delivered into heating/melting/vaporizing the electrode metal. The same heat balance can also be applied to energy or heat (Q), as to power (P). It can also be seen that $k_p = P_p/P_{EI}$. In accompanying experiments, the radiated fraction of energy was found to vary with the electrode type. After the initial transient, about 60% of the energy appeared as radiation in the case of steel electrodes, 45% for aluminum, and 30% for copper. The authors considered that, with the copper electrodes, the low radiant fraction was a measurement error, since only a finite spectral window of radiation could be measured, and that the true value would be around 50%, in which case a 100% heat balance would achieved.

Using some approximate calculations [39], the Aachen researchers concluded that about 10 – 12% of the energy went into heating,

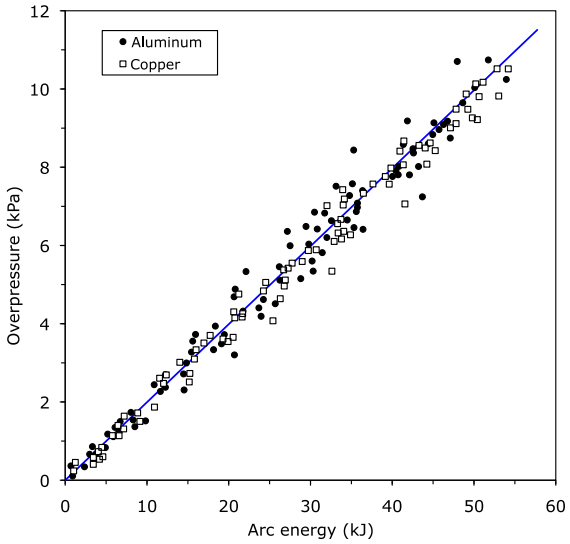


Fig. 6. Relation between arc energy and pressure, as found by RWTH Aachen University [32] for a 1000 V capacitive discharge in an 0.8 m^3 enclosure using 120 – 150 mm arc gaps.

Table 1
Theoretical heat losses into electrodes.

Metal	Heat required to gasify (MJ kg^{-1})	Heat required to oxidize (MJ kg^{-1})	Net heat absorbed by metal (MJ kg^{-1})
Al	13.73	−30.0	−16.3
Cu	6.18	−1.1	5.0
Fe	8.04	−4.9	3.7

melting, or vaporizing the electrodes. The authors also did some measurements on open arcs, not within enclosures, and obtained essentially identical values for the thermal radiation fraction. From this, they concluded that for partially-vented enclosures, it would be possible to estimate the expected pressure rise by a subtraction process, i.e., subtracting the radiation loss and the electrode loss from 100%. Comparable data published by Thai researchers [40] indicate that 16% of the electrical arc energy went into heating Cu electrodes, in the case of a phase-to-phase fault. But in the case of a fault from phase to cabinet, 17% of the energy went into electrode heating, with an additional 12% going into heating the steel of the cabinet. Such results would not be applicable to high-current short arcs, where the vaporization of the metal becomes more important [41]. A theoretical treatment was provided by Binnendijk et al. [42] (Table 1), but such theoretical numbers would not be achieved in practice, since only a fraction of the metal can possibly get oxidized. Nonetheless this indicates that aluminum electrodes are likely to behave differently from copper or steel, in that heat gains from oxidation will offset some of the heat extracted due to gasification.

These preceding studies suggest that there *should* be a difference between aluminum and copper electrodes, despite the Aachen results (Fig. 6) not showing this effect. Tanaka et al. [43] however successfully documented that a difference does exist between aluminum, copper, and steel electrodes. Fig. 7 shows their results obtained for AC arcs of fixed 0.1 s duration. The computed values of k_p then form an ordered series (0.86, 0.54, and 0.31) which reflects the energy gains found with aluminum that do not occur with copper or steel.

In some other experiments [39], the Aachen group found that with 2.8 MJ of arc energy, a peak overpressure of 2 atm was obtained. These measurements, however, were in a test rig where a pressure relief valve opened after the initial part of the pulse, thus a quasi-steady-state

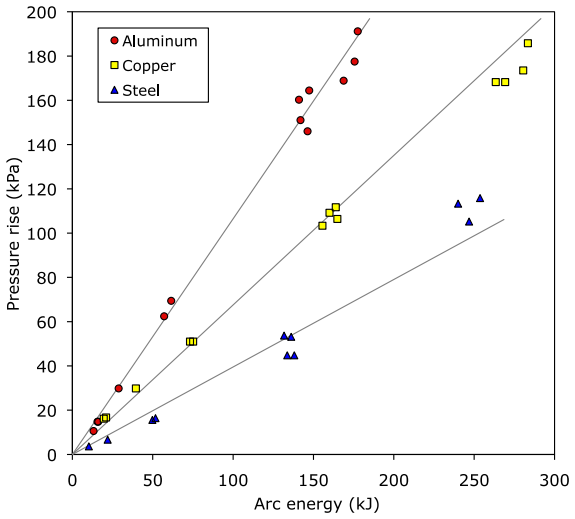


Fig. 7. Arc pressure rise, as function of electrode material, measured by Tanaka et al. [43] in an 0.32 m^3 enclosure for 0.1 s duration AC arcs.

result was not obtained. In further experiments [44], also using relief-vented compartments, they showed that, unlike for energy, there is no specific relationship between peak arc *power* and the overpressure. Raising the arc power from 2.6 to 7.8 MW only changed the overpressure in these tests from 1.1 to 1.3 atm.

The Aachen treatment is simple, but explicitly ignores *time* effects, which can be major. Baldrey and Hudson[33] ran some time-resolved tests, where they found that, at the inception of arcing, nearly 100% of the arc energy went into heating the air (and thereby raising the pressure), but after three cycles (60 ms), the value dropped to 12% and was still trending downwards. Conversely, the fraction of energy going into heating or vaporizing electrodes steadily rose. Aachen researchers also found later that the factor k_p is not a constant of a particular experiment, but varies with the current level [45]. For copper electrodes, the current-dependence was insignificant, but with aluminum and steel electrodes k_p rose monotonically with current, although not linearly. Concerning the time effect in general, most of the published results for enclosures have involved some form of pressure relief. In such cases, the pressure drops rapidly after the vent opens and both the peak pressure and the time to the peak are largely controlled by the vent characteristics. There is so little information on unvented, unruptured enclosures that it is difficult to make any generalizations concerning the time characteristics of the pressure envelope.

When experiments comparing SF_6 to air atmospheres were run [39], it was found that a higher fraction of the arc energy was recovered as a pressure rise in the gas, typically 80%, but ranging from 60% to 90%. The heat balance here is not clear, since, with as much as 90% of the arc energy going into raising the gas pressure, it would leave only trivial amounts for thermal radiation and for electrode heating. Interestingly, a French research group [46] did a combination of modeling and experiments with SF_6 and concluded that only 10% of the arc energy will go into raising the gas pressure. They did not do any experiments with air, but in comparing arcs in N_2 and in SF_6, found that, for the same arc current, the pressure rise in SF_6 is very close to 50% of what it is with N_2. Further work at Aachen [47] produced the relationship:

$$k_p = 0.6\frac{c_v}{c_{v(SF_6)}}$$

where k_p = fraction of energy going into raising the pressure and the temperature of the gas, c_v = heat capacity for gas of interest, and $c_{v(SF_6)}$ = heat capacity of SF_6. This is based on assuming that k_p =0.60 for SF_6. Thus, if c_v =88.97 J mol^{-1} K^{-1} for SF_6 and c_v =20.88 J mol^{-1} K^{-1} for air, then for air k_p =0.14. For N_2, c_v

=20.81 J mol^{-1} K^{-1}, giving k_p =0.14 also. But according to the French research k_p for SF_6 is closer to 0.10 than to 0.60. Thus, the proposed estimating rule should be viewed with caution. It also needs to be pointed out that the Aachen researchers ignored all other heat effects and assumed that all of the arc power is partitioned solely between radiation and heating/pressurizing the gas. For air, this would then indicate that 86% of the energy gets radiated away. Thus, this does not even agree with the earlier Aachen results, where for tests in air, 45 – 60% of the energy went into radiation, depending on the electrode type. Also, the proposed relationship has no electrode-material dependence, while their experimental results showed a significant effect.

The Aachen group also ran tests [48] to explore the effect of enclosure volume and electrode material on k_p using two enclosures, of 0.07 and 0.14 m^3 volume. Using both air and SF_6, aluminum electrodes gave higher k_p values overall, but additionally, in the case of aluminum electrodes, k_p was higher for the smaller enclosure, while for copper electrodes k_p was higher for the larger enclosure. For copper electrodes, there was relatively little change in the enclosure pressure due to vaporization of copper, but for aluminum electrodes, this factor became sizable. In air, there is a compensating factor in that substantial oxidation of Al vapors occurred, lowering the pressure due to formation of condensed-phase products, while the SF_6 atmosphere is inert and oxidation does not take place. With aluminum electrodes and the 0.07 m^3 enclosure, the pressure rise was actually greater than for the larger enclosure, since there was insufficient oxygen available to react with enough of the Al vapors in order to lower the pressure. These results are consistent with the theoretical calculations Table 1 but raise the question as to why copper and aluminum results are essentially indistinguishable in Fig. 6. It would appear possibly this is due to a fortuitous choice of enclosure size in those tests.

Efforts have been made to use commercial CFD codes for design purposes [56], but the validity of doing this is an unsettled matter. The Aachen researchers [44] compared several 1-d and 2-d models for predicting arc pressure and found modest agreement, except from flow through openings. The models did not encompass acoustic waves and could not predict the actual heat balance from first principles, so the researchers had to provide a value of k_p as input data. In a later study [49], they used 3-d CFD modeling which included temperature-dependent gas properties; this gave excellent predictions, with both experiment and calculation showing a peak overpressure of 3 atm. They also developed a simplified CFD 'ray-tracing' procedure [50], but this was seen to be more limited than standard 3-d CFD modeling. Other efforts have not met with much success. Capelli-Schellpfeffer et al. published two CFD modeling studies intended to simulate an experimental arc fault in 480 VAC switchgear in which an arc energy of 20 kJ was delivered and a peak overpressure of 2.6 atm was measured [51]. In the first study [51], a peak of 0.2 atm was computed, while in the second [52] a peak of 16,000 atm was predicted. Interestingly, the same 20 kJ value of arc energy was used by Caillard et al. [53] but in a capacitive discharge circuit and they predicted a peak overpressure of 1700 atm while measuring a peak of 0.26 atm.

In case of a switchgear enclosure which vents into a limited-size outer room, a two-compartment model must be considered. Pietsch et al. [39] performed such calculations and reported some experimental results. Even et al. [54] reported some modeling studies on a 3-compartment arrangement done with a CFD code, where the middle compartment readily sustained overpressures of 22 kPa. Schamale and Pietsch [55] also reported similar calculations and obtained up to 0.9 atm overpressures in the compartment of origin.

5. Acoustic measurements

Any arc will cause pressure excursions in the air which will be perceived as sound when the frequencies are in the range the human ear can hear, 20 Hz – 20 kHz. If the discharge is in a closed compartment, this will additionally excite standing waves which are characteristic of the shape and dimensions of the enclosure. For example, Wahle and Summer [56] ran arcing tests in a commercial switchgear cabinet and obtained the pressure waveform shown in Fig. 8. The power supply was 50 Hz, but a high frequency component was roughly 1500 Hz, which presumably corresponds to one of the acoustic standing-wave modes of the cabinet. In general, inside an enclosure the peak overpressure when taking acoustic waves into account may be significantly higher than the pressure calculated from a quasi-steady-state solution. A simplified estimating technique for this has been proposed [50], but it has not been validated.

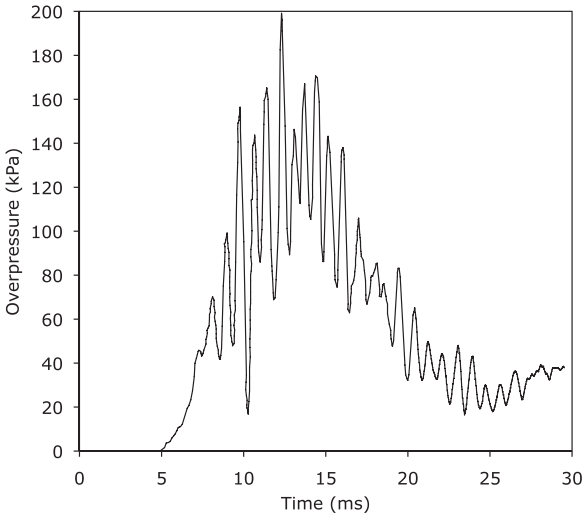


Fig. 8. Arc pressure measured in a switchgear cabinet by Wahle and Summers [56].

However, a closed compartment is not needed for high-frequency oscillations to be found. Even for a capacitive discharge arc taking place in open air, where there is no powerline frequency involved, the discharge circuit may generate a complex spectrum. Nettleton [57] showed that such arcs exhibit a primary frequency, characteristic of a circuit's dominant LC resonance, overlaid by a high-frequency component. Such circuit-related oscillations have not been studied in any substantive way. There is however yet a third source of frequencies found in the sound spectrum of an arc explosion, associated with the actual shock wave generation process. Dawson et al. [58] studied this process and found the dominant frequency to be $f = c\sqrt{p/W}$, where c = speed of sound (m s^{-1}), p= ambient pressure (Pa), and W = energy per length of gap (J m^{-1}).

In early studies on this topic, McFarlane [59] measured the sound waves from capacitive discharge arcs and concluded that the sound pressure level (SPL) varies with distance as $1/r$, while the acoustic energy flux ('sound intensity'; power per unit area) varies as $1/r^2$. These scaling relations are consistent with the general theory of acoustics. Suits [18] described that, very roughly, the fraction of the arc energy going into producing waves in the air is comparable to the fraction lost as radiation; while Flowers [19] proposed a model for the acoustic energy output of an arc in open air. He considered that the simplest model is one where the sound energy produced is equal to the displacement of air by the expanding arc channel, as given by:

$$\Delta Q = \frac{1}{2} m v^2$$

where ΔQ = acoustic energy (J), m = mass of air displaced (kg), and v = expansion velocity (m s^{-1}). The mass of air displaced is given by $m = \rho_o A_a L$, where ρ_o = ambient density of air (kg m^{-3}), A_a = arc channel cross-section (m^2), and L = arc length (m). As discussed above, Flowers had determined that the arc channel cross-section is directly proportional to arc current; however, he did not offer a theory to predict v, and only provided some experimental data, which showed values were typically 1000 – 2000 m s^{-1}.

Lee[7] reported "arcs sounded like a sizeable artillery piece firing" at a distance of 1 mile (1.6 km) in tests using a 230 kV source out in the open, but this did not lead to persons getting blown down at a distance of 15 m. Jones et al. [60] reported making SPL measurements and documenting pressures that exceeded 185 dB (35 kPa) during small, "nonevent" explosions. They considered this problematic since such pressures would rupture eardrums, however, they did not publish details of their work. Doughty et al. [61] measured 3-phase arc explosions where the arc currents ranged up to 29 kA; they recorded a peak SPL of 163 dB (2.8 kPa). In general, it can be concluded that acoustic phenomena have been studied only to a limited extent, SPL measurements of arc explosions are rare, and available guidance is scant.

6. Summary and conclusions

When an arc breakdown is initiated, energy gets deposited into the arc channel at a rate much greater than can be removed from the area by the shock wave that is created. This causes a rapid pressure rise and, if the arc energy is sufficiently high, this will be perceived as an explosion. For a low-energy arc, the perceived sound may simply be a 'snap,' 'crackle,' or 'pop.' But within the scientific community there is not an agreed-upon, quantitative definition[14] of the term 'explosion,' nor are there studies to quantify the fraction of the arc energy that gets delivered as sound energy, i.e., vibrations in the 20 Hz – 20 kHz range.

Electric arc explosions are not combustion phenomena—they are predominantly physical explosions, due to very rapid conversion of electrical energy into heat. Chemical reactions play a role, but only a supporting role, in such explosions. Recent studies suggest that chemical reactions are mainly ones which convert air to species such as O_3 and NO_x. While these may comprise oxidation, they are very different from a fuel-air explosion of a normal sort. Some of the electrode metal is also vaporized in an arc explosion, and arc temperatures are high enough so that presumably much of this metal vapor may get oxidized. However, the electrode oxidation effect is quantitatively only a small part of the heat balance in an arc explosion. Thus, first-order estimates of arc explosions treat the process solely as converting electrical energy to heat and ignore chemical reaction contributions.

In an open environment, arc pressures will rarely be highly destructive. Theoretical modeling suggests that very high pressures may be created, but experimental measurements at practical distances do not suggest a high overpressure hazard. Using a 4 MV discharge, Uman[27] did not measure any overpressures above 10 kPa, while Drouet and Nadeau's work [30] indicates that at 0.3 m from an 480 V_{rms}, 80 kA_{rms} arc, an overpressure of only 8.7 kPa will result. These values are all well below levels at which eardrum rupture may be anticipated.

But on the other hand, if a high fault current is available and arcing takes place in a small enclosure, extremely high overpressures can be reached, measured in one case at 409 atm. Arc flash, which is not considered in this paper, comprises the main injury potential in open arcing, instead of high pressures. For arcing in small enclosures, the initial hazard may be destruction of equipment. But serious injuries may occur as consequential damages, e.g., trauma to persons from airborne cabinet parts. In addition, arcing may continue after a cabinet failure and injuries from arc flash may then ensue.

Most accidents involve AC power. However, for arcing within small enclosures, DC arcs are more hazardous than capacitive discharge or AC arcs. With DC arcs, arcing can continue for a long time, and during this time the enclosure pressure will progressively rise. With capacitive discharge arcs, the event is intrinsically short-lived. With AC arcs, continued re-ignitions of the arc may take place, nonetheless peak overpressures will tend to be much lower than for DC conditions. A direct, quantitative comparison between DC and AC arcs is not possible, however, since AC arcs do not have a unique waveform and will depend on circuit characteristics. While many accidents involving AC arcs take place where voltages are 480 VAC, or higher, destructive arcing can also occur with 120 – 240 VAC sources, provided sufficient current capacity, e.g., kiloamps, is available.

Very little test or modeling data exist on arcing in closed, unvented, unruptured enclosures. This would be especially germane to arcs where the power source is sustained, i.e., DC or AC arcs, as opposed to capacitive discharges. These conditions may not necessarily represent a direct injury potential, but the analysis is simpler and reliable data would useful for validating of theories. Apart from some very early studies, there has not been significant experimental work on DC arcs. A number of industries use DC and studies specific to DC power supplies would be valuable.

Even in the simplest case of an arc explosion—an explosion in open air—the existing research base is inadequate. The primary collection of results are from Drouet and Nadeau [30], but their results show some anomalies which place in doubt the accuracy of their work. It must further be cautioned that Drouet and Nadeau's study was popularized by the 1987 paper of R.H. Lee [7], which introduced major errors not present in the original.

Electrodes can be of diverse metals, with copper, aluminum and steel being the most common. The literature is conflicted on the effect that electrode metal type is likely to make, and more focused research would be desirable.

In line with other fields of engineering, in recent years it has become common to conduct research studies based on CFD modeling. Unfortunately such studies only rarely lead to phenomenological insights or conclusions having general applicability. In addition, unless accompanied by a careful and successful validation, the numerical results may be in serious error. And the CFD studies reported thus far on arc explosions have generally not been successful in the validation efforts against experimental data.

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